Session 18 · Tuesday 27 October 2026 · One derivation. Start wherever the scaffolding stops helping you.

The same cycle, three times

A protein that exists already, and the only thing changing is whether a phosphate is on it. One kinase pushing right, one phosphatase pushing left:

$$\text{forward} = \frac{V_1 W}{K_{m1} + W}, \qquad \text{back} = \frac{V_2 W^*}{K_{m2} + W^*}, \qquad W + W^* = W_T.$$

Two tools, and nothing else. **Steady state is equal and opposite flux**, not zero flux — both enzymes run flat out and the pool sits still. And **the total is conserved**, which is the first time in this course that anything has been.

Throughout, $w = W^*/W_T$ is the fraction modified, and

$$k_1 = \frac{K_{m1}}{W_T}, \qquad k_2 = \frac{K_{m2}}{W_T}.$$

Those two are the whole session. They are dimensionless, they are **not** properties of the enzymes, and a bigger pool makes them **smaller**.

At each transition the sheet asks *why does that step follow?* Answer in writing before you go on.

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1 · Fully worked — where the curve crosses one half

Set the fluxes equal.

$$\frac{V_1 W}{K_{m1} + W} = \frac{V_2 W^*}{K_{m2} + W^*}$$

Put the conservation law in. $W = W_T(1 - w)$ and $W^* = W_T w$, so every numerator carries a factor of W_T:

$$\frac{V_1 W_T (1 - w)}{K_{m1} + W_T(1 - w)} = \frac{V_2 W_T\, w}{K_{m2} + W_T w}$$

Divide through by W_T. It cancels from the numerators and divides into each K_m downstairs, leaving k_1 and k_2 behind:

$$\frac{V_1 (1 - w)}{k_1 + 1 - w} = \frac{V_2\, w}{k_2 + w}$$

Divide by V_2. Only the ratio survives — five parameters are now three:

$$\frac{(V_1/V_2)(1 - w)}{k_1 + 1 - w} = \frac{w}{k_2 + w}$$

Now set $w = 1/2$ and ask what ratio puts it there:

$$\frac{(V_1/V_2)\cdot\frac{1}{2}}{k_1+\frac{1}{2}}=\frac{\frac{1}{2}}{k_2+\frac{1}{2}}\qquad\Longrightarrow\qquad\left(\frac{V_1}{V_2}\right)_{50}=\frac{k_1+\frac{1}{2}}{k_2+\frac{1}{2}}$$

▶ **Why does that step follow?** W_T divided out of the equation. Explain, in one sentence, why the pool size nevertheless still matters.

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2 · Partly worked — the one point every response curve passes through

Now ask a different question. Sweep V_1/V_2 at several different saturations and you get a family of curves — and they all meet at a single point, at $V_1/V_2 = 1$. Find the w at which they cross, as a function of k_1 and k_2 only.

The condition. Put $V_1/V_2 = 1$ into the quadratic. The w^2 term disappears, because its coefficient is $(1 - V_1/V_2)$ — which is why this is the one stimulus where the model has an exact answer. Show it.

The answer, which you should reach:

$$w=\frac{k_2}{k_1+k_2}=\frac{K_{m2}}{K_{m1}+K_{m2}}$$

▶ **Why does that step follow?** W_T appears in both k_1 and k_2 and cancels in the ratio, so the crossing point does not move when you change the pool size — at all, exactly. Say in one sentence what that means for a designer who wants to shift *where* the switch sits.

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3 · Scaffolding gone — how sharp, and in what units

A cycle has $k_1 = k_2 = k$ and you may vary V_1/V_2.

(a) Goldbeter and Koshland's ruler: R is the V_1/V_2 needed to reach $w = 0.9$, divided by the V_1/V_2 needed to reach $w = 0.1$. Work out R for a plain Michaelis–Menten curve $y = x/(1 + x)$. You should get a pure number, the same for every K.

(b) A Hill curve of coefficient n scores $81^{1/n}$ on that ruler. Invert it to get n in terms of R.

(c) For $k = 10^{-2}$ the model gives $R \approx 1.18$. What apparent Hill coefficient is that? Compare it with hemoglobin, which is about 3.

(d) Nothing in this cycle binds cooperatively. In one sentence: where does the sharpness come from?

▶ **Why does that step follow?** A sharper switch has a **smaller** R. Explain why the ruler runs that way round.

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4 · The ledger — what you would actually turn

You have three things you can change: the expression ratio V_1/V_2, the two Michaelis constants K_{m1} and K_{m2}, and the pool size W_T.

YOU WANT TO MOVE	WHICH ONE, AND WHY
where the switch crosses	
how sharp it is	
where it fires	

(e) Two of those three answers are a promoter and one is protein engineering. Which is which?

(f) One pair of goals cannot both be had. Name the pair, and say what the trade is.

(g) The cycle costs ATP continuously, even when nothing is changing and the output is not moving. Why — and what does that tell you about when **not** to use this design?

Before Thursday

Mishra, Bepler, Teague, Berger, Broach & Weiss, *Science* **373** (2021) — a toggle switch with no promoter in it, built from eleven phospho-in, phospho-out elements. Come able to put their network on session 8's axes: activity of one branch across, activity of the other up, the two nullclines drawn, every intersection marked, and a statement about which are stable.

Their Fig. 4C is the control that earns the result. Work out what it is controlling for before you read what they say about it.